THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The sum of the first 35 odd numbers is 1225. The sum of the odd numbers from 31 to 69 is: 1

(a) 1000 **(b)** 1210 **(c)** 400 **(d)** 969

2. A spinner starts pointing North and turns clockwise 315° then anticlockwise 45°. The final angle clockwise from North is: 1

(a) 280 **(b)** 360 **(c)** 270 **(d)** 260

3. The number of 4-digit palindromes with first digit 5 is: 1

(a) 90 **(b)** 100 **(c)** 9 **(d)** 10

4. A bar graph has scale 1 cm = 15 units. The heights of two bars differ by 6 cm. The values differ by: 1

(a) 21 **(b)** 90 **(c)** 105 **(d)** 75

5. The number of composite numbers from 1 to 50 (both included) is: 1

(a) 36 **(b)** 34 **(c)** 35 **(d)** 33

6. A square and an equilateral triangle have equal perimeters. If the square's side is 15 cm, the triangle's side (cm) is: 1

(a) 35 **(b)** 20 **(c)** 15 **(d)** 25

7. A drum is $\frac{5}{6}$ full. 24 litres are removed and it becomes $\frac{1}{3}$ full. The capacity (litres) is: 1

(a) 47 **(b)** 49 **(c)** 48 **(d)** 58

8. A circle of radius 5 cm is drawn. A square of side equal to the diameter is drawn around it. The area of the square (cm^2) is: 1

(a) 95 **(b)** 110 **(c)** 105 **(d)** 100

9. After a rotation of 200° a regular 9-gon coincides with itself. This rotation is 5 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 50 **(b)** 40 **(c)** 70 **(d)** 30

10. The sum of the integers from -8 to 12 (both included) is: 1

(a) 43 **(b)** 42 **(c)** 52 **(d)** 41

11. The difference between the 25th and the 24th square numbers is: 1

(a) 47 **(b)** 51 **(c)** 49 **(d)** 25

12. The smaller angle between the hands of a clock at 2 o'clock is 60°. The smaller angle between the hands at 10 o'clock is: 1

(a) 120 **(b)** 90 **(c)** 60 **(d)** 300

13. The sum of 8 consecutive counting numbers starting from 25 is: 1

(a) 264 **(b)** 200 **(c)** 236 **(d)** 228

14. A school bus carries 40 children on trip 1, 55 on trip 2 and 35 on trip 3. A pictograph uses one symbol for 5 children. The number of symbols needed for all three trips is: 1

(a) 28 **(b)** 27 **(c)** 26 **(d)** 25

15. The smallest number divisible by each of 10, 15 and 20 is: 1

(a) 59 **(b)** 61 **(c)** 60 **(d)** 70

16. A rectangular field 30 m by 16 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 210 **(b)** 270 **(c)** 256 **(d)** 240

17. A rod 9 m long is cut into pieces of $\frac{3}{5}$ m each. The number of pieces is: 1

(a) 14 **(b)** 16 **(c)** 15 **(d)** 17

18. Two circles of equal radius 6 cm touch externally. The distance from the outer end of one to the outer end of the other along the line of centres (cm): 1

(a) 30 **(b)** 18 **(c)** 36 **(d)** 24

19. The number of letters among O, X, D, M, T with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4

20. On a number line, the distance between -7 and -23 is: 1

(a) 16 **(b)** 19 **(c)** 17 **(d)** 15

21. A pattern adds the same number every time. Its first term is 12 and its 17th term is 220. The number added each time is: 1

(a) 12 **(b)** 13 **(c)** 15 **(d)** 14

22. The minute hand of a clock takes how many minutes to turn through 240°? 1

(a) 45 **(b)** 8 **(c)** 40 **(d)** 20

23. A 5-digit number ends with digit 4 and has digit sum 28; the sum of its other four digits is: 1

(a) 15 **(b)** 33 **(c)** 32 **(d)** 24

24. Out of 200 students, 85 play cricket and 64 play football, none plays both. How many play neither? 1

(a) 61 **(b)** 115 **(c)** 51 **(d)** 136

25. The greatest number that divides 60, 90 and 150 leaving no remainder is: 1

(a) 31 **(b)** 30 **(c)** 32 **(d)** 29

26. A picture 40 cm by 25 cm is mounted on a frame of width 4 cm all around. The area of the frame alone (cm^2) is: 1

(a) 580 **(b)** 584 **(c)** 600 **(d)** 588

27. If $\frac{2}{7}$ of a number is 8, then $\frac{3}{8}$ of the number is: 1

(a) 8 **(b)** $10\frac{1}{2}$ **(c)** $1\frac{1}{2}$ **(d)** 28

28. In rectangle ABCD, AC = BD = 12 cm. If O is the intersection, the sum OA + OB + OC + OD (cm) is: 1

(a) 18 **(b)** 36 **(c)** 30 **(d)** 24

29. The sum of the angles of rotation of a square and a regular octagon (in degrees) is: 1

(a) 125 **(b)** 135 **(c)** 145 **(d)** 165

30. Two integers add to 8 and one of them is -9. The other is: 1

(a) 19 **(b)** 17 **(c)** 15 **(d)** 27

31. A pattern adds 1, then 2, then 3, ... Starting at 25, the term after 15 additions is: 1

(a) 40 **(b)** 145 **(c)** 250 **(d)** 130

32. 5 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 4 **(b)** 6 **(c)** 15 **(d)** 5

33. The number of digits in the largest palindrome smaller than 1000 is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

34. A bar graph shows sales of ₹1800, ₹1200, ₹1600. If the scale is 1 cm = ₹200, the total height of all three bars is (cm): 1

(a) 25 **(b)** 4600 **(c)** 23 **(d)** 21

35. For the 3-digit number 2d5, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 3 **(b)** 4 **(c)** 11 **(d)** 2

36. A wire in the shape of a square of side 25 cm is straightened and bent into a rectangle of length 32 cm. The breadth (cm) is: 1

(a) 11 **(b)** 7 **(c)** 32 **(d)** 18

37. Which is the smallest of $\frac{4}{5}$, $\frac{11}{15}$, $\frac{7}{9}$, $\frac{3}{4}$? 1

(a) $\frac{4}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{11}{15}$

38. Three circles of radius 5 cm are drawn in a triangle arrangement, each touching the other two. The triangle joining the centres has perimeter (cm): 1

(a) 35 **(b)** 30 **(c)** 25 **(d)** 45

39. A figure has rotational symmetry of order 5 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 7

40. A trader loses ₹80 on day 1, gains ₹60 on day 2 and loses ₹90 on day 3. The net result in ₹ is: 1

(a) -100 **(b)** -112 **(c)** -110 **(d)** -108

41. **Assertion (A):** The sum $1 + 2 + 3 + \cdots + 10$ is 55. 1
Reason (R): The sum of the first n natural numbers is $n \times n$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** The minute hand of a clock turns through 90° in 15 minutes. 1
Reason (R): The minute hand turns through 30° in every minute.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** A Kaprekar step on 5432 gives 5432 - 2345 = 3087. 1
Reason (R): A Kaprekar step adds the largest and the smallest arrangements of the digits.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** In a bar graph with scale 1 cm = 20 students, a bar 4 cm tall shows 80 students. 1
Reason (R): To find the number from a bar, divide the bar height by the scale value.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** 91 is a composite number. 1
Reason (R): Every number that is not divisible by 2, 3 or 5 is a prime number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** A rectangle has area 48 cm^2 and length 8 cm. Its breadth is 6 cm. 1
Reason (R): Breadth = area x length.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$. 1
Reason (R): To add fractions with the same denominator, add the numerators and also add the denominators.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** A point 5 cm from the centre of a circle of radius 4 cm lies inside the circle. 1
Reason (R): A point whose distance from the centre is less than the radius lies inside the circle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A parallelogram that is not a rectangle or rhombus has rotational symmetry of order 2. 1
Reason (R): Such a parallelogram has two lines of symmetry.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** $-4 + 6 = 2$. 1
Reason (R): Subtracting a negative integer is the same as subtracting the corresponding positive integer.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**